Project Title -: **Virtual Travel Assistant**

Name – Vartika

Admission No – 25SCS1003004515

Batch – 1CSE31

Faculty Name – Dr. Lalita Chaudhary , AP , SCSE IILM University , Greater Noida

Problem Context : - Many of the tourists face struggle regarding lack of information including local attractions and hidden gems. They find difficulty in planning a trip especially to unfamiliar places. People shifts from websites to videos to blogs researching about the best weather to visit the location, cost required for the trip, affordable hotels.

Problem Definition : - An AI based travel assistant which will be intelligent, centralized and proactive platform for the entire travel journey. It will make travel journey very easy and smooth.

Objectives :-

The project aims to achieve the following objectives :

1. Personalized Travel Planning- Provide customized tour plan based on user preferences (budget, interests, duration, location).

2. Intelligent Destination Recommendation- Use ML models to recommend destinations, activities, and accommodations using user history, reviews, and trends.

3. Real Time Assistance - Offer live travel information such as weather, flight/train status, and local transport availability.

4. Natural Language Interaction- Enable users to interact with the assistant through text or voice, using NLP for queries like “Suggest me a 3-day trip under ₹20,000.”

5. Smart Budget Optimization - Suggest cost-effective routes, accommodations, and packages to fit within the user’s financial constraints.

Input : -

1. Personal Preferences- Budget, Travel Date, Interests etc.
2. User Profile History- Past Trips, Frequently visited travel pages, Frequently Searched Destinations

3. Language Selection - Preferred Language for Conversation.

Output : -

- Personalized travel recommendations
- Improved adaptive recommendations
- Responds in users selected language

Scopes : -

- Provide live weather updates, transport status, and safety alerts during travel.
- Allow interaction in multiple languages for better understanding .
- Suggest budget-friendly routes, hotels, and transport options.

Limitations : -

- Needs good connectivity of internet.
- Suggestions may not always reflect hidden charges (taxes, local fees, etc.).

Real Life Relevance : -

- Simplifies travel planning- Tourists often take too much time browsing on different websites but virtual Travel Assistant make this process easy which saves time and effort.

- Real Time Support- Help travelers to find live updates on weather, transport delays and safety alerts.

- Budget Friendly- It provides the users best deals on flights, stays and local activities.

- Email ID - vartika.25scs1003004515@iilm.edu

- Phone No - 8307546544